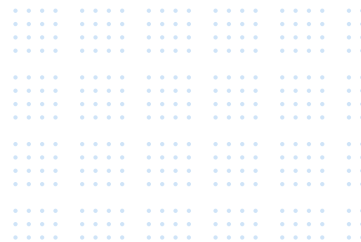


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Afghan International Islamic University

Faculty of Computer Science
Master of Science in Artificial Intelligence

ADVANCED PROGRAMMING FOR AI



PRACTICAL ASSIGNMENT

NumPy, Pandas, and Web Scrapping for AI Data Preparation

Course Code: **M.AI.1116**

Instructor

Dr. Ziaullah Momand

Assignment Type

Individual-based practical
assignment

Total Marks

100 marks

Student Name: _____

Student ID: _____

Submission Date: _____

Semester: Spring 2026

NumPy, Pandas, and Web Scraping for AI Data Preparation

Master-level practical assignment | Total Marks: 100 | Assignment Type: Individual-based

1. Assignment Title

Building an AI-Ready Dataset Using Web Scraping, External Datasets, NumPy, and Pandas.

2. Assignment Description

In this assignment, students will collect data from the web using web scraping and combine it with another external dataset. They will clean, transform, analyze, and prepare the data using Pandas and NumPy.

The focus of this assignment is data preparation, not model training. Students should not train a machine learning model for this assignment.

Students must demonstrate that they can:

1. Scrape structured data from a website.
2. Load another dataset from an external source.
3. Clean and prepare both datasets.
4. Use Pandas for filtering, grouping, merging, sorting, and feature engineering.
5. Use NumPy for numerical computation and AI-related data representation.
6. Produce a final clean dataset that could later be used for AI or machine learning work.

3. Learning Outcomes

After completing this assignment, students should be able to:

1. Collect data from the web using Python.
2. Extract meaningful information from HTML pages.
3. Load CSV, JSON, or Excel datasets using Pandas.
4. Clean missing, duplicate, and inconsistent data.
5. Merge, combine, or compare datasets from different sources.
6. Use Pandas for data inspection, filtering, grouping, sorting, and transformation.
7. Use NumPy arrays for numerical computation.
8. Create an AI-ready numerical feature matrix from cleaned data.
9. Document the complete data preparation workflow clearly.

4. Dataset Requirements

Students must use at least two datasets:

- Dataset 1: a web-scraped dataset collected using Python.
- Dataset 2: an external dataset loaded from a public or instructor-approved source.

Dataset 1: Web-Scraped Dataset

Students must scrape data from a website using Python. Recommended practice websites are shown below.

Option A: Books Dataset

<https://books.toscrape.com/>

Possible fields: book title, price, rating, availability, category, and product link.

Option B: Quotes Dataset

<https://quotes.toscrape.com/>

Possible fields: quote text, author, tags, author profile link, number of tags, and quote length.

Option C: Custom Website

- The website must contain structured data.
- It must not require login.
- It must not contain private or sensitive data.
- It should allow educational use.
- The website URL must be mentioned in the report.

Dataset 2: External Dataset

In addition to web scraping, students must use another dataset from an external source. The second dataset should be related to the web-scraped dataset as much as possible.

- Kaggle dataset.
- UCI Machine Learning Repository dataset.
- Public GitHub CSV dataset.
- Government open data portal.
- University open dataset.
- Public JSON dataset.
- Any instructor-approved CSV, JSON, or Excel file.

Dataset Type	Example
Books	Book ratings, authors, genres, prices
Education	Student performance, attendance, exam records
Health	Public health statistics or non-personal hospital data
Economy	Country GDP, population, unemployment
Sports	Player statistics or team performance
Movies	Movie ratings, genres, release years
E-commerce	Product prices, reviews, ratings
Social Data	Public quotes, topics, tags, categories

Web-Scraped Dataset	External Dataset
Books from Books to Scrape	Public books dataset from Kaggle
Quotes from Quotes to Scrape	Public authors or quotes dataset
Product prices from a website	Public product review dataset
Country information from a website	Country population or GDP CSV
Movie data from a website	IMDb-style movie dataset

5. Main Assignment Tasks

Part A: Web Scraping and Data Collection - 20 Marks

Students must write Python code to scrape data from a selected website.

Required Tasks

1. Import the required libraries.
2. Send an HTTP request to the selected website.
3. Parse the HTML content using BeautifulSoup.
4. Extract at least 5 meaningful fields.
5. Scrape at least 100 records if the website supports pagination.
6. Store the scraped records in a Pandas DataFrame.
7. Save the raw scraped dataset as scraped_raw_dataset.csv.

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np

url = "https://quotes.toscrape.com/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

records = []
# Extract data and append dictionaries into records

df_scraped = pd.DataFrame(records)
df_scraped.to_csv("scraped_raw_dataset.csv", index=False, encoding="utf-8")
```

Example Field for Books	Example Value
Title	A Light in the Attic
Price	GBP 51.77
Rating	Three
Availability	In stock
Product Link	catalog/a-light-in-the-attic_1000/index.html

Example Field for Quotes	Example Value
Quote	The world as we have created it...
Author	Albert Einstein
Tags	change, deep-thoughts, thinking
Author Link	/author/Albert-Einstein
Number of Tags	3

Part B: Loading an External Dataset - 10 Marks

Students must load a second dataset from another source. The dataset must be relevant to the web-scraped dataset or useful for comparison.

1. Select a relevant external dataset.
2. Mention the dataset source in the report.
3. Load the dataset using Pandas.
4. Inspect the dataset structure.
5. Save a copy of the original external dataset as external_raw_dataset.csv.

```
# For CSV

df_external = pd.read_csv("external_dataset.csv")

# For JSON

df_external = pd.read_json("external_dataset.json")
```

```
# For Excel

df_external = pd.read_excel("external_dataset.xlsx")

# Inspect the external dataset

df_external.head()
df_external.shape
df_external.info()
df_external.describe()
df_external.isna().sum()

df_external.to_csv("external_raw_dataset.csv", index=False, encoding="utf-8")
```

Part C: Pandas Data Cleaning and Preparation - 25 Marks

Students must clean both datasets using Pandas. Cleaning decisions must be explained in the notebook and report.

1. Inspect Both Datasets

```
df.head()
df.tail()
df.shape
df.info()
df.describe()
df.columns
df.dtypes
df.isna().sum()
```

- Explain the number of rows and columns.
- Identify the data types of important columns.
- Identify missing values and duplicate records.
- Explain which columns are important for analysis.

2. Handle Missing Values

```
df.isna().sum()

# Option 1: remove rows with missing values

df = df.dropna()

# Option 2: fill missing text values

df["column_name"] = df["column_name"].fillna("Unknown")

# Option 3: fill missing numerical values

df["column_name"] = df["column_name"].fillna(df["column_name"].median())
```

Students must explain why they selected each missing-value handling method.

3. Remove Duplicate Records

```
df.duplicated().sum()
df = df.drop_duplicates()
```

This must be applied to both the scraped dataset and the external dataset.

4. Clean Text Columns

```
df["title"] = df["title"].str.strip()
df["author"] = df["author"].str.strip()
df["quote"] = df["quote"].str.replace('\"', '')
```

Students should remove unnecessary spaces, symbols, or inconsistent text formatting.

5. Convert Data Types

```
# Example for price

df["price"] = df["price"].str.replace("GBP", "").astype(float)

# Example for rating

rating_map = {
    "One": 1,
    "Two": 2,
    "Three": 3,
    "Four": 4,
    "Five": 5
}

df["rating_numeric"] = df["rating"].map(rating_map)
```

6. Filtering

Students must apply at least 3 filtering operations.

```
df_high_rating = df[df["rating_numeric"] >= 4]
df_available = df[df["availability"] == "In stock"]
df_expensive = df[df["price"] > 40]
```

7. Sorting

Students must sort the data based on at least 2 meaningful columns.

```
df.sort_values("price", ascending=False)
df.sort_values(["category", "rating_numeric"], ascending=[True, False])
```

8. Grouping and Aggregation

Students must apply at least 3 grouping operations.

```
df.groupby("category")["price"].mean()
df.groupby("rating_numeric")["title"].count()
df.groupby("availability")["price"].agg(["mean", "min", "max"])
```

9. Feature Engineering

Students must create at least 4 new features.

```
# Examples for books

df["title_length"] = df["title"].str.len()
df["is_expensive"] = df["price"] > df["price"].mean()
df["price_level"] = pd.cut(
    df["price"],
    bins=[0, 20, 40, 100],
    labels=["Low", "Medium", "High"]
)
df["rating_level"] = df["rating_numeric"].apply(
    lambda x: "High" if x >= 4 else "Low"
)

# Examples for quotes

df["quote_length"] = df["quote"].str.len()
df["word_count"] = df["quote"].str.split().str.len()
df["number_of_tags"] = df["tags"].str.split(",").str.len()
df["is_long_quote"] = df["word_count"] >= 20
```

Part D: Combining or Comparing Datasets - 15 Marks

Students must use both datasets together in a meaningful way. There are two acceptable approaches.

Option 1: Merge the Datasets

If both datasets have a common column, students should merge them.

```
merged_df = pd.merge(
    df_scraped,
    df_external,
    on="author",
    how="inner"
```

```
)

# Other possible merge types

pd.merge(df_scraped, df_external, on="column_name", how="left")
pd.merge(df_scraped, df_external, on="column_name", how="outer")
```

- Explain which column was used for merging.
- Explain which merge type was used.
- Report how many rows existed before and after merging.
- Explain whether missing values were created after merging.

Option 2: Compare the Datasets

If the datasets cannot be merged directly, students should compare them using similar features.

Web-Scraped Dataset	External Dataset	Possible Comparison
Scraped book prices	External book prices	Compare average prices
Scraped book ratings	External book ratings	Compare rating distributions
Scraped quotes	External quotes dataset	Compare quote length and tag count
Scraped countries data	External population data	Compare country-level statistics

Students should produce at least 3 comparison results using Pandas.

```
df_scraped["price"].mean()
df_external["price"].mean()

df_scraped.groupby("category")["price"].mean()
df_external.groupby("category")["price"].mean()

df_scraped["rating_numeric"].value_counts()
df_external["rating_numeric"].value_counts()
```

Part E: NumPy for AI Computing - 15 Marks

Students must use NumPy to perform numerical operations on the cleaned dataset.

1. Convert Selected Columns to a NumPy Array

```
numeric_data = df[["price", "rating_numeric", "title_length"]].to_numpy()

# Or for quotes

numeric_data = df[["quote_length", "word_count", "number_of_tags"]].to_numpy()

numeric_data.shape
```

- Explain the shape of the array.
- Explain the number of rows and columns.
- Explain the meaning of each numerical column.

2. Basic NumPy Statistics

```
np.mean(numeric_data, axis=0)
np.median(numeric_data, axis=0)
np.std(numeric_data, axis=0)
np.min(numeric_data, axis=0)
np.max(numeric_data, axis=0)
```

Students should explain the meaning of each result.

3. Vectorized Computation

Students must perform at least 2 vectorized operations.

```
df["price_usd"] = df["price"] * 1.27

df["normalized_price"] = (
    df["price"] - np.mean(df["price"])
```

```
) / np.std(df["price"])
```

4. Normalization or Standardization

Students must normalize or standardize at least 2 numerical columns.

```
# Min-max normalization

df["price_normalized"] = (
    df["price"] - df["price"].min()
) / (df["price"].max() - df["price"].min())

# Standardization

df["price_standardized"] = (
    df["price"] - df["price"].mean()
) / df["price"].std()
```

Students must explain why normalization is useful for AI data preparation.

5. Feature Matrix Representation

```
X_matrix = df[["price", "rating_numeric", "title_length"]].to_numpy()
print(X_matrix.shape)
```

In the feature matrix, **rows represent samples** and **columns represent features**. The purpose is only to represent data in numerical matrix form.

Part F: Final AI-Ready Dataset Preparation - 5 Marks

Students must save the final cleaned and prepared dataset. The final dataset should include cleaned original columns, engineered features, numerical columns useful for AI, no unnecessary duplicates, and properly handled missing values.

```
df.to_csv("ai_ready_dataset.csv", index=False, encoding="utf-8")
```

Students should briefly describe why the dataset is ready for future AI-related analysis.

Part G: Report Writing - 10 Marks

Students must submit a short report with the following structure.

Section	Required Explanation
1. Introduction	Assignment purpose, selected web-scraped dataset, selected external dataset, and reason for selection.
2. Web Scraping Process	Website URL, scraped fields, number of records, pagination method, and challenges.
3. External Dataset Description	Dataset source, file format, rows, columns, main fields, and relevance.
4. Data Cleaning	Missing values, duplicates, text cleaning, and data type conversion.
5. Pandas Analysis	Filtering, sorting, grouping, aggregation, and engineered features.
6. Dataset Combination or Comparison	Merge or comparison approach, common columns, similarities, differences, and problems faced.
7. NumPy Analysis	Array shape, statistics, vectorized operations, normalization, and feature matrix representation.
8. Conclusion	What was learned, main challenges, and how Pandas and NumPy support AI data preparation.

6. Final Submission Requirements

File	Description
assignment.ipynb	Full Jupyter or Google Colab notebook
scraped_raw_dataset.csv	Raw dataset collected through web scraping
external_raw_dataset.csv	Original external dataset
clean_scraped_dataset.csv	Cleaned scraped dataset

File	Description
clean_external_dataset.csv	Cleaned external dataset
ai_ready_dataset.csv	Final prepared dataset
report.pdf or report.docx	Short explanation report

7. Marking Rubric

Component	Marks
Web scraping implementation	20
External dataset loading and explanation	10
Pandas cleaning and preparation	25
Dataset merging or comparison	15
NumPy numerical computing	15
Final AI-ready dataset	5
Report quality and explanation	10
Total	100

8. Detailed Rubric

A. Web Scraping - 20 Marks

Criteria	Marks
Correct use of requests and BeautifulSoup	4
Extracts at least 5 useful fields	4
Handles pagination or collects sufficient records	4
Stores scraped data in a DataFrame	4
Saves raw scraped dataset as CSV	4

B. External Dataset - 10 Marks

Criteria	Marks
Selects a relevant external dataset	3
Loads dataset correctly using Pandas	3
Inspects dataset structure clearly	2
Explains dataset source and relevance	2

C. Pandas Cleaning and Preparation - 25 Marks

Criteria	Marks
Dataset inspection using Pandas functions	3
Missing value handling	4
Duplicate removal	3
Text cleaning and formatting	3
Data type conversion	4
Filtering and sorting	3
Grouping and aggregation	3
Feature engineering	2

D. Dataset Combining or Comparison - 15 Marks

Criteria	Marks
Correctly merges or compares datasets	5
Uses meaningful columns or features	4
Explains the relationship between datasets	3
Interprets results clearly	3

E. NumPy - 15 Marks

Criteria	Marks
Converts DataFrame columns to NumPy array	3
Explains array shape and structure	3
Computes numerical statistics	3
Uses vectorized operations	3
Applies normalization or standardization	3

F. Final AI-Ready Dataset - 5 Marks

Criteria	Marks
Saves final dataset correctly	2
Dataset is clean, structured, and usable	3

G. Report - 10 Marks

Criteria	Marks
Clear explanation of data sources	2
Clear explanation of scraping process	2
Clear explanation of cleaning process	2
Clear explanation of Pandas and NumPy analysis	2
Clear conclusion and formatting	2

9. Academic Integrity and Ethical Rules

1. Do not copy another student's notebook.
2. Do not manually copy web data.

3. Web data must be collected using Python code.
4. The external dataset source must be clearly mentioned.
5. Do not scrape login-protected websites.
6. Do not collect private, personal, or sensitive information.
7. Do not overload websites with repeated requests.
8. Use scraping only for educational and legal purposes.
9. The submitted notebook must run from beginning to end.
10. All code must be properly commented.

10. Suggested Notebook Structure

1. Assignment Title and Student Information
2. Import Libraries
3. Web Scraping
 - 3.1 Send request
 - 3.2 Parse HTML
 - 3.3 Extract fields
 - 3.4 Handle pagination
 - 3.5 Store scraped data in DataFrame
 - 3.6 Save scraped raw dataset
4. External Dataset Loading
 - 4.1 Load CSV/JSON/Excel dataset
 - 4.2 Inspect external dataset
 - 4.3 Save external raw dataset
5. Data Cleaning
 - 5.1 Clean scraped dataset
 - 5.2 Clean external dataset
 - 5.3 Handle missing values
 - 5.4 Remove duplicates
 - 5.5 Convert data types
 - 5.6 Clean text columns
6. Pandas Analysis
 - 6.1 Filtering
 - 6.2 Sorting
 - 6.3 Grouping
 - 6.4 Aggregation
 - 6.5 Feature engineering
7. Dataset Combining or Comparison
 - 7.1 Merge datasets if possible
 - 7.2 Otherwise compare similar features
 - 7.3 Interpret results
8. NumPy Analysis
 - 8.1 Convert numerical columns to NumPy array
 - 8.2 Check array shape
 - 8.3 Compute statistics
 - 8.4 Perform vectorized operations
 - 8.5 Normalize or standardize numerical columns
 - 8.6 Create feature matrix
9. Final AI-Ready Dataset
 - 9.1 Select final columns
 - 9.2 Save final dataset
10. Conclusion

11. Example Assignment Question

You are required to build an AI-ready dataset using Python. First, scrape structured data from a suitable website. Second, load another related external dataset from a public source such as Kaggle, UCI, GitHub, or an open data

portal. Then use Pandas to inspect, clean, filter, group, sort, transform, and combine or compare the datasets. After that, use NumPy to perform numerical operations, normalization, vectorized computation, and feature matrix representation.

Your final output should include raw datasets, cleaned datasets, a final AI-ready dataset, a complete Jupyter or Google Colab notebook, and a short report. **You are not required to train any machine learning model.**

12. Expected Final Output Example

For a books-based assignment, the final dataset may look like this:

title	price	rating_nu meri c	title_length	price_level	is_expensi ve
A Light in the Attic	51.77	3	21	High	True
Tipping the Velvet	53.74	1	18	High	True
Soumission	50.10	1	10	High	True
Sharp Objects	47.82	4	13	High	True

Possible NumPy feature matrix:

```
[[51.77, 3, 21],  
 [53.74, 1, 18],  
 [50.10, 1, 10],  
 [47.82, 4, 13]]
```

- Rows represent data samples.
- Columns represent numerical features.
- The dataset is clean and ready for future AI-related analysis.

13. Final Note for Students

In real AI projects, data preparation is one of the most important stages. Before using any AI or machine learning method, the data must be collected, cleaned, structured, transformed, and represented properly. This assignment gives practical experience with three essential tools: web scraping for data collection, Pandas for data preparation, and NumPy for numerical computing.